Chapter #13

Genetic Algorithm

1. Explain the main steps of Genetic algorithm with illustrative example?

Definition:

- A genetic algorithm is an adaptive heuristic search algorithm inspired by Darwin's theory of evaluation in nature.
- Solve complex problem that take very long time.
- Solve optimization problems

General characteristics of GA

⇒ Derivative free
⇒ stochastic optimization / randomness
⇒ Inspired by natural selection
⇒ Proposed by John Holland in 1975.

Components:

Population:

Subset of all possible or probable solutions, which can solve given problem.

Chromosomes:

A chromosome is one of the solution in the population for the given problem.

Gene:

A chromosome is divided into different genes or it is an element of the chromosomes.

Allele:

Allele is the value provided to the gene within a particular chromosome.

Fitness function:

To determine the individual fitness level in the population. means the ability of an

individual to compete with other individual.

In every iteration individuals are evaluated based on their fitness function.

Genetic Operators

The best individuals made to regenerate offspring better than parents.

Genetic operator play a role in changing the genetic composition of the next generation.

Selection:

a selection process is used to determine which individualities in the population will get to reproduce and produce the seed that will form coming generation.

Selection styles

- Event Selection
- Rank-grounded
- Roulette wheel

DATE: / /20

How genetic algorithm works?

The GA works on the evolutionary generational cycle to generate high quality solutions.

It basically involves fine steps.

1) Initialization:

⇒ Process ~~starts by~~ generating the set of individuals, called population.

⇒ each individual is the solution for the given problem.

⇒ An individual contains or characterized by set of parameters called genes.

⇒ Genes are combined into string and generate chromosomes which is the solution to the problem.

⇒ one of the popular initialization technique is the use of random binary string.

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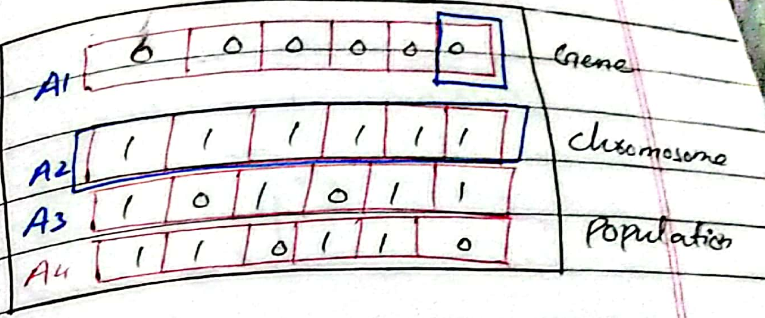


2) Fitness assignment:

⇒ Function used to determine how fit and individual is?

⇒ Ability of an individual to compete with other individuals.

⇒ In each iteration individuals are evaluated based on their fitness function.

⇒ it provides fitness score of each individual.

⇒ based on this score individuals are selected for further reproduction.

⇒ Having high score have more chance for being selected.

3) Selection:

⇒ This phase involves the selection of individuals of offspring.

⇒ All selected individuals are then arranged in pair of two to increase reproduction.
Then these individuals transfer their genes to the next generation.

Types of selection method

- Rank-based
- Tournament
- Roulette wheel

4) Reproduction:

⇒ After selection, the creation of child occurs in the reproduction step.

⇒ In this step, two variation operators used that are applied to the parent population.

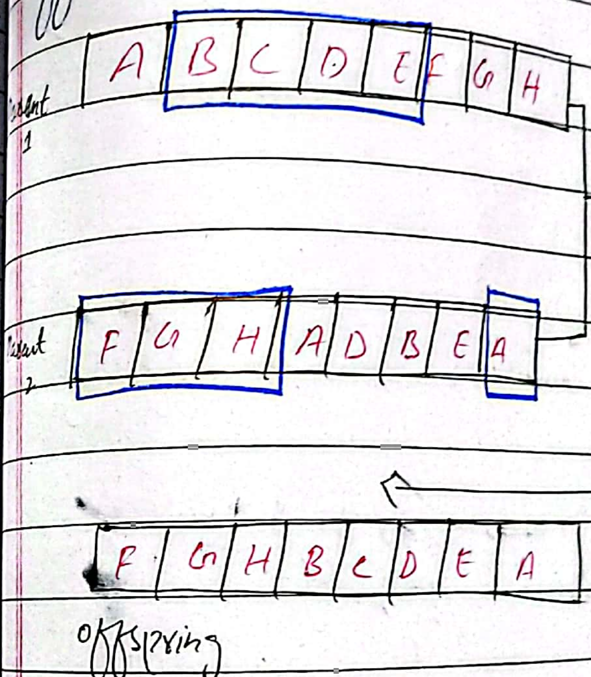
Two operators are:

Cross over:

Play significant role.

a cross point is selected randomly in gene.

Then crossover operator swaps genetic information of two parents from the current generation to produce a new individual representing the offspring.



DATE: __/__/20

⇒ The genes of parents are exchanged among themselves until the crossover point is met.

⇒ Newly generated child are added to the population.

⇒ This process is called the crossover.

Types of crossover

- one point
- Two point
- Linear
- inheritance

mutation:

⇒ The mutation operator inserts random genes in the offspring (new child) to maintain the diversity in population.

⇒ Can be done by flipping some of the bits in the chromosomes:

⇒ Solve premature convergence issues.

DATE: __/__/20

Types of mutation

- flip bit
- Gaussian
- swap/exchange

F	G	H	B	C	D	E	A	Before mutation
F	G	M	B	C	D	E	N	After mutation

5) **Termination:**

⇒ After the reproduction phase, a stopping criteria is applied as a base for termination.

⇒ Stop algorithm after the threshold fitness solution is met.

⇒ final solution is the best solution in the population

Pros:

- No need for derivatives
- Population based search
- probabilistic nature
- parallel implementation

Cons:

- Not efficient for solving simple problems.
- Does not give guarantee of final solution.
- Repetitive calculation of fitness values may generate some computational challenge.

Characteristics of GA:

Parallel search:

Can be implemented on parallel-processing machine

Versatility:

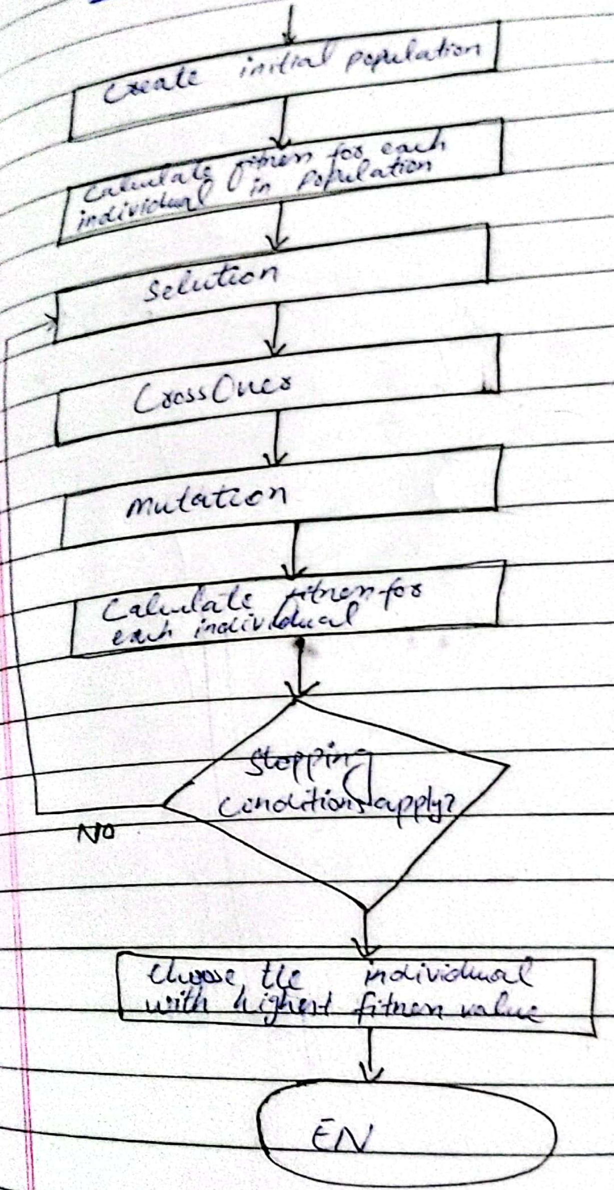
Applicable to both continuous and discrete optimization problem.

Avoid local minima:

Stochastic nature of GA helps avoid getting trapped in local minima.

Flexibility:

Flow chart



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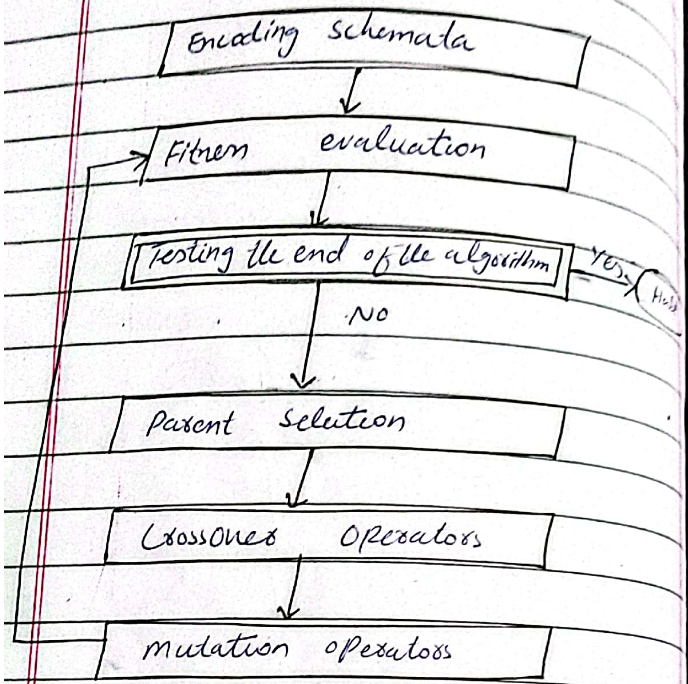
Major Phases



Define schemata?

a schema (plural of schemata) is a template that defines a subset of chromosomes in a search space that share similarities.

DATE : __/__/20

Comparison of GA with Natural evolution:

Natural evolution	Genetic Algo
Chromosome	String
Gene	feature in string
Locus	Position in string
Allele	Position value
Genotype	String structure
Phenotype	Set of characteristics